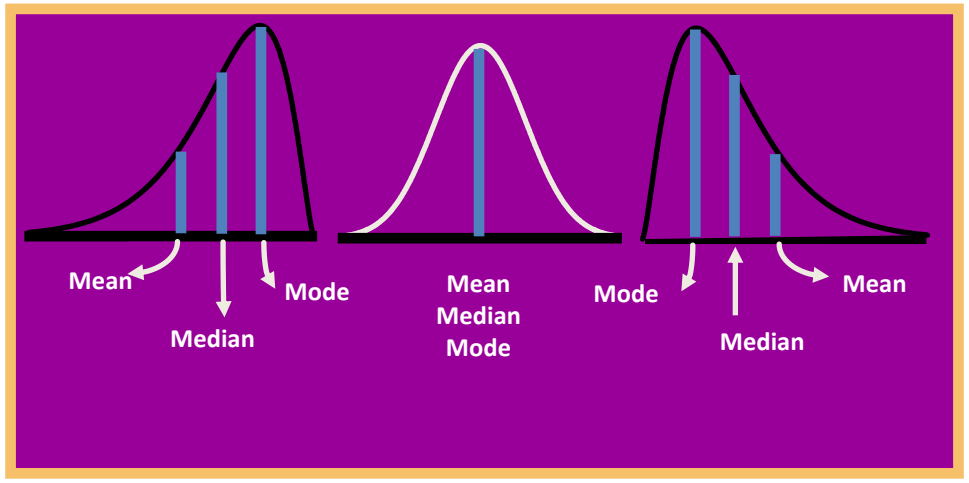
Shape of the distribution curve

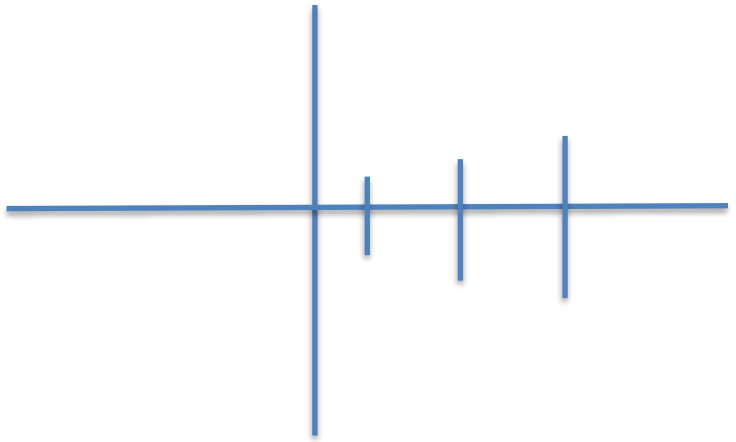
Chapter 2 (Part B)

Biostatistics and Data Analysis



Contents

- Shape of the distribution
- Skewnes
  - ✓ Positive skew
  - ✓ Negative skew
  - ✓ Normal distribution
  - ✓ Quantile Definition of Skewness

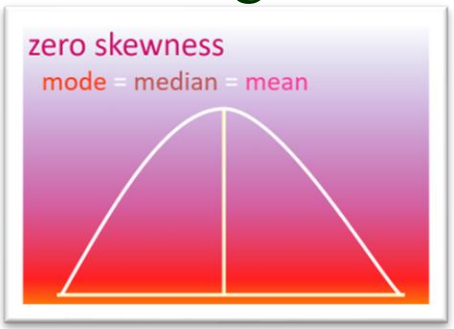


SHAPE OF THE DISTRIBUTION

Shape of the Distribution

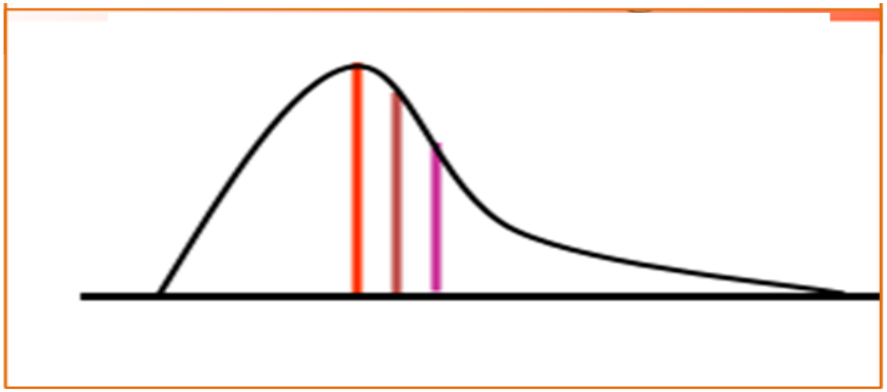
Symmetric Distribution

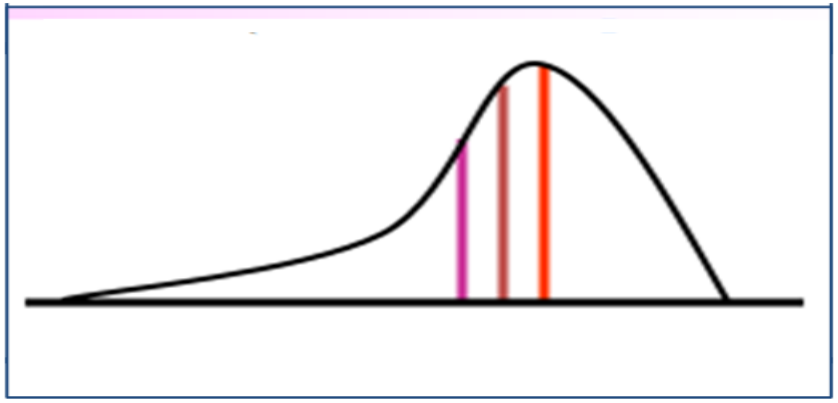
- The patterns radiating from the central point, on the left and right side are same to each other.



Non-Symmetric Distribution

- The patterns radiating from the central point, on the left and right side are different from each other.





- **Skewness**
  - ✓ **Positive skew**
  - ✓ **Negative skew**
  - ✓ **Normal distribution**
  - ✓ Quantile Definition of Skewness

Skewness

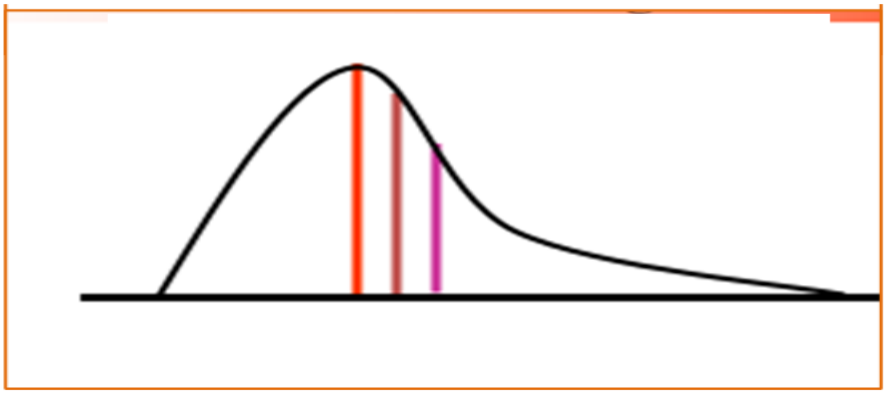
- When the patterns radiating from the central point, on the left and right side are different from each other, it is termed as **Skewness**

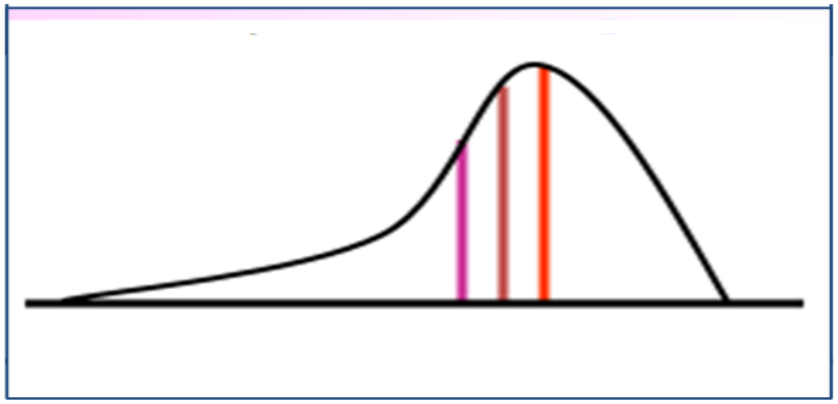
OR

- Extreme values on one side of a distribution.

OR

- Absence of symmetry.





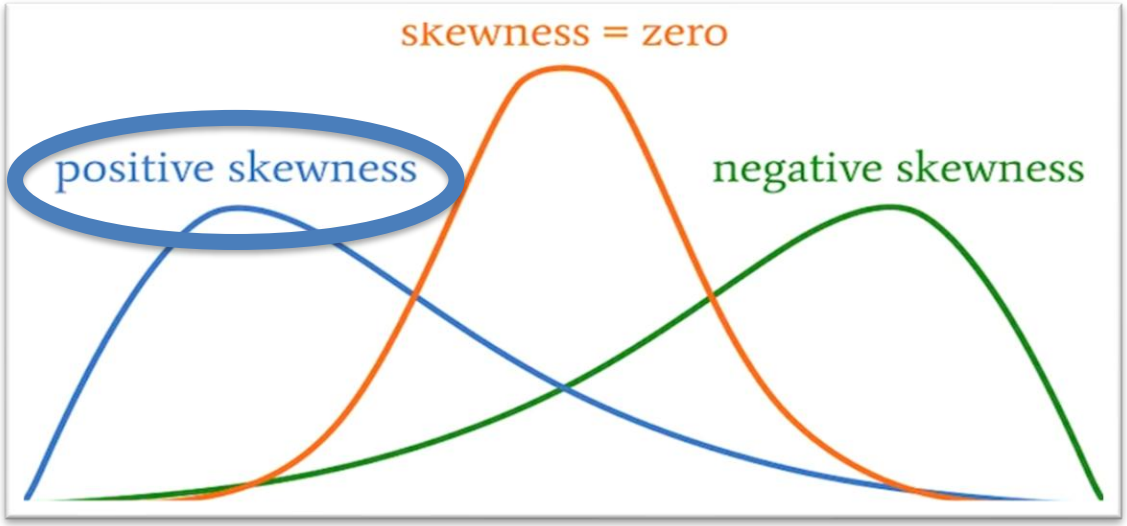
Skewness

Positively Skewed Distribution

- In a **positively skewed distribution**, most data values **cluster on the left side** (lower values).
- The **tail extends to the right**, showing a few **larger or extreme values**.

📈 **In summary:**

- The scores pile up on the **left**, and the **right tail is longer**.



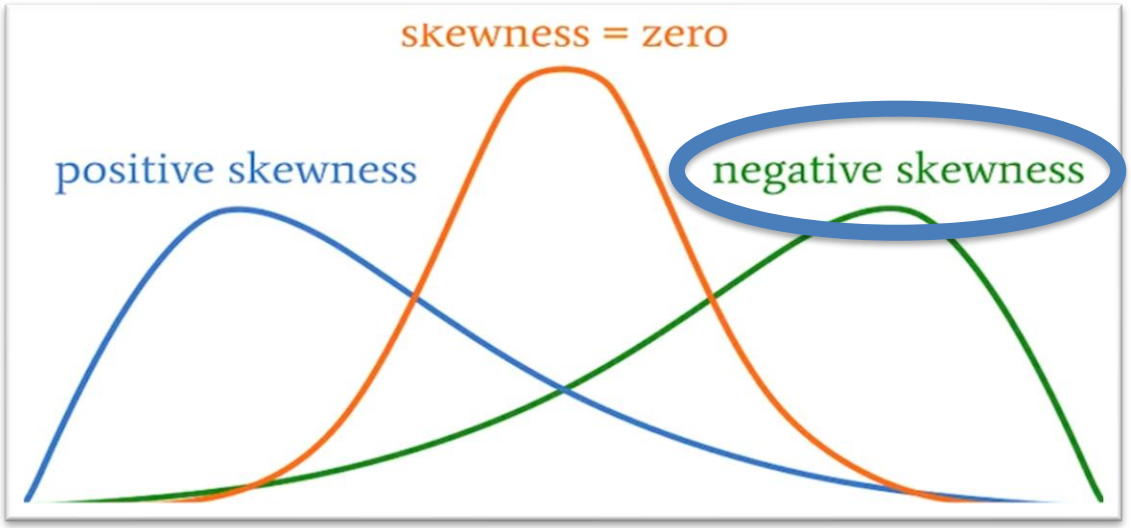
Skewness

Negatively Skewed Distribution

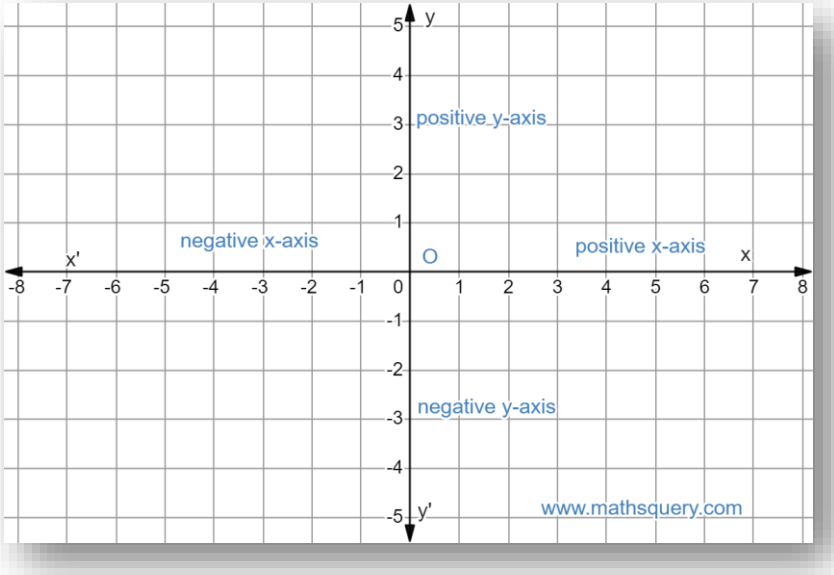
- In a **negatively skewed distribution**, most data values **cluster on the right side** (higher values).
- The **tail extends to the left**, showing a few **smaller or extreme low values**.

📈 **In summary:**

- The scores pile up on the **right**, and the **left tail is longer**.



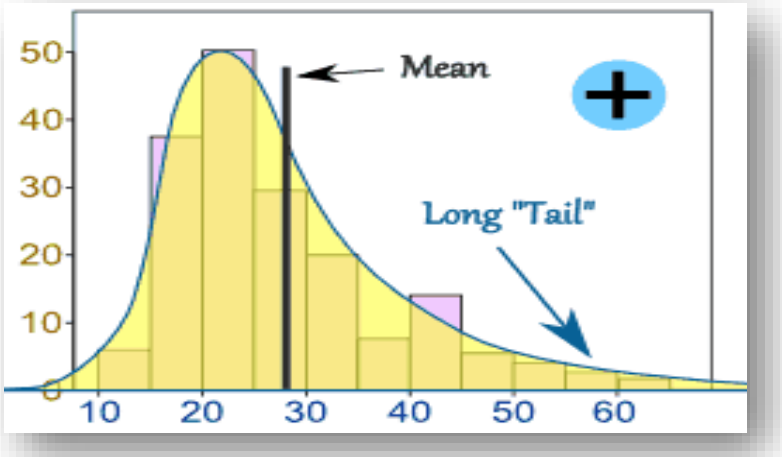
Skewness



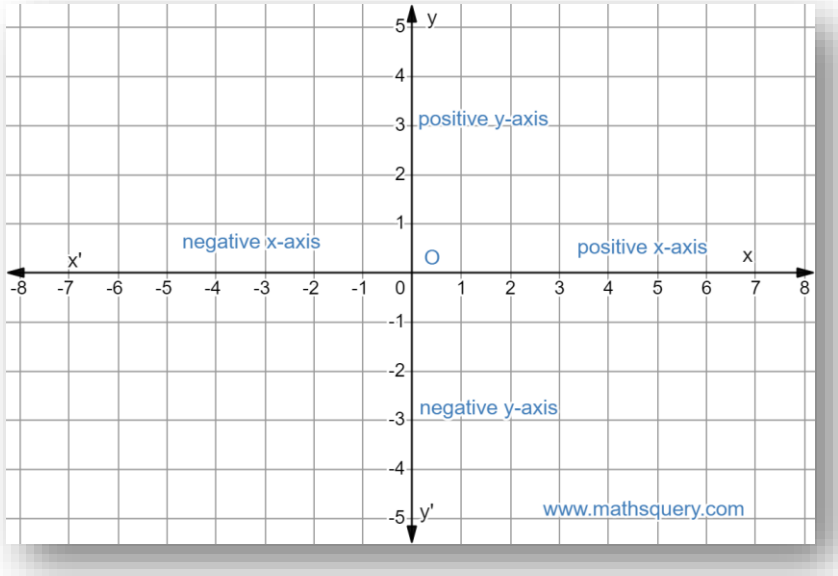
Positively Skewed Distribution

Why is it called **positive** skew?

- Because the long "tail" is on the positive side of the peak or the long tail is on the right-hand side.
- Tail extends to the right (positive x-axis).



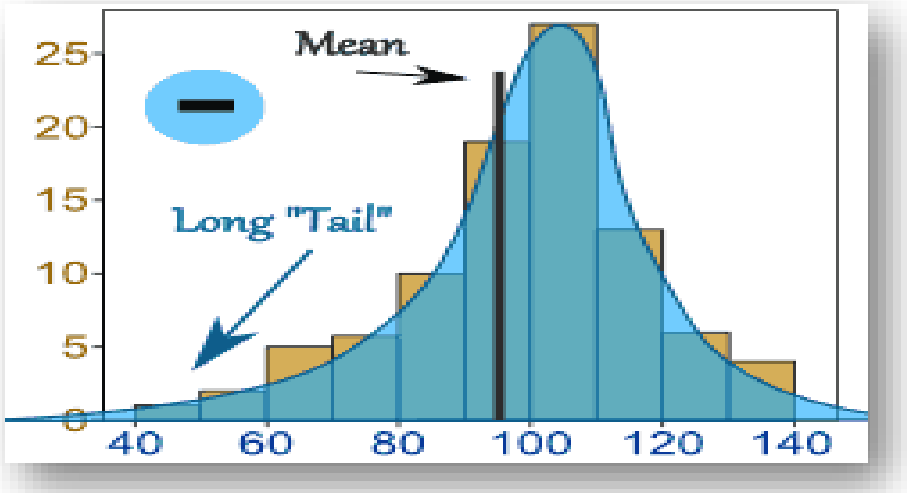
Skewness

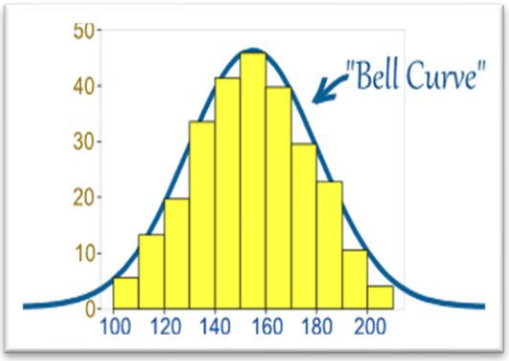


Negatively Skewed Distribution

Why is it called **negative** skew?

- Because the long "tail" is on the negative side of the peak or the long tail is on the left-hand side.
- Tail extends to the left (negative x-axis).





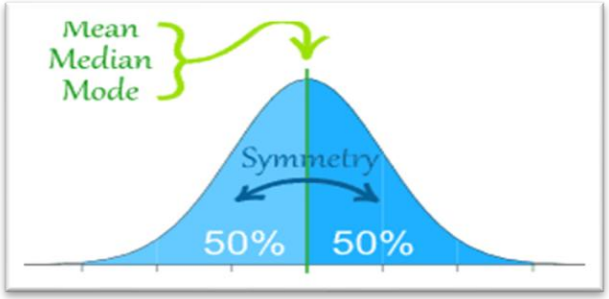
Skewness

Zero/No-Skewed

Normal Distribution

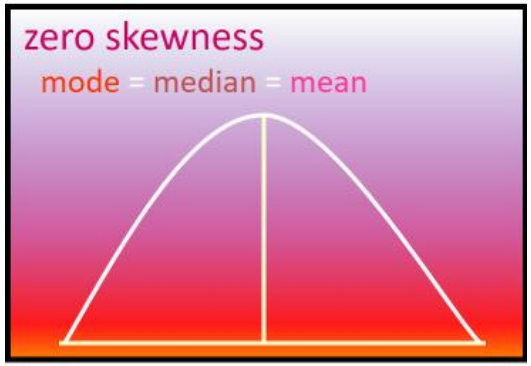


- It is also known as the Gaussian Distribution and the curve is also known as the Gaussian Curve, named after German Mathematician Carl Frederich Gauss.
- The Normal Distribution has Bell-shaped curve.
- A Normal Distribution is not skewed.
- It is perfectly symmetrical.
- Two halves of the curve are the same (mirror images).
- The Mean is exactly at the peak.
- 50% of the values less than the mean and 50% greater than the mean.

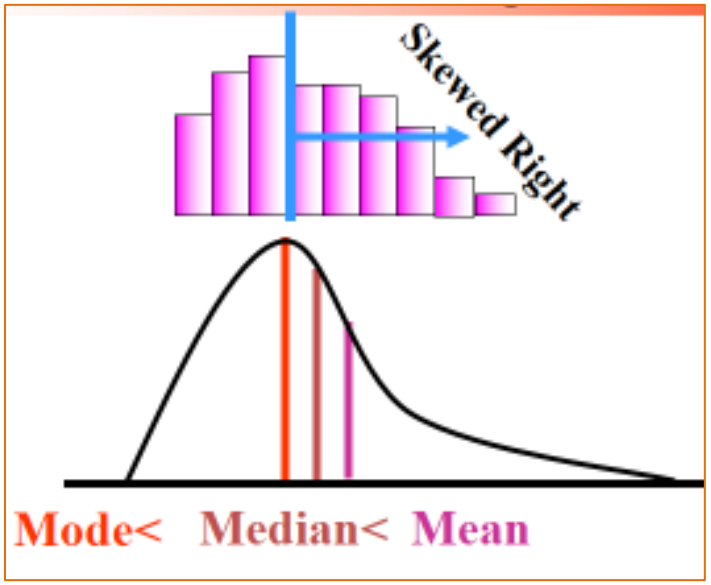


Skewness

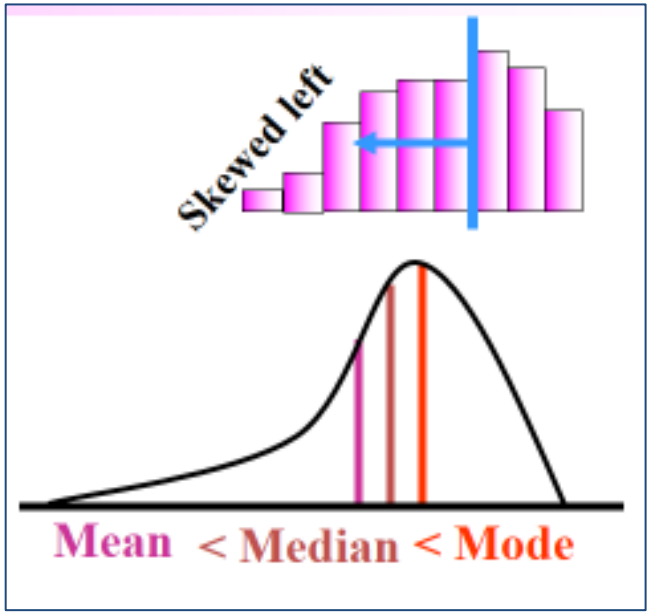
Positive or Negative or Zero/No-Skewed



Positively skewed

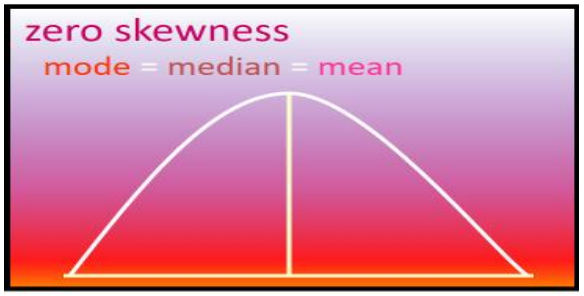


Negatively skewed



- In symmetrical/normal distributions (Zero/No Skewed)

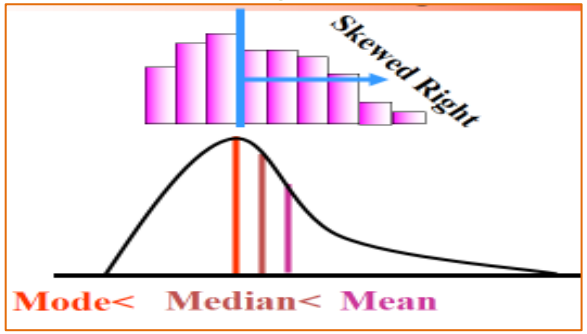
Mean = Median = Mode



- In **positively skewed distributions**, the mean is greater than the median and mode.

Mode <Median < Mean

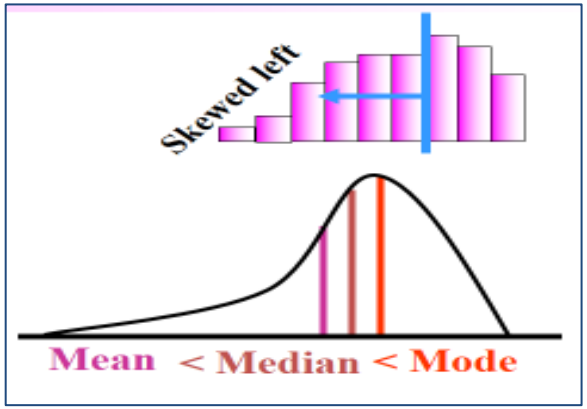
Positively skewed



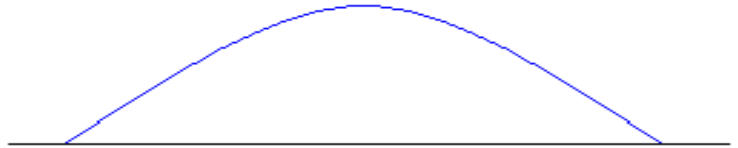
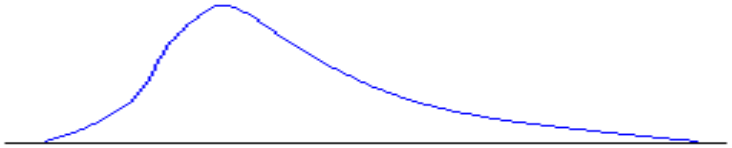
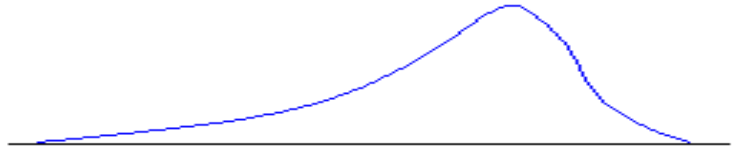
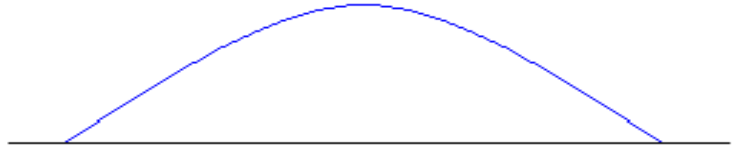
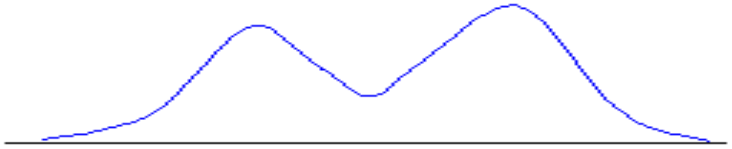
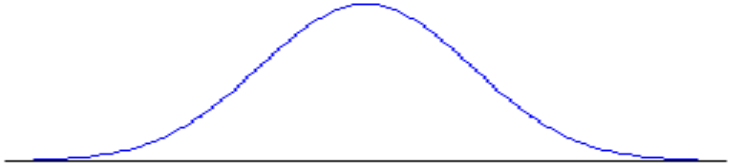
- In **negatively skewed** distributions, the mean is smaller than the median and mode.

Mode >Median>Mean

Negatively skewed



Terms that Describe Distributions

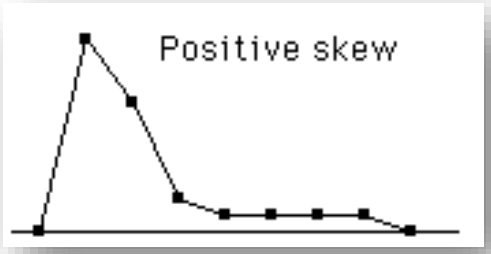
Term	Features	Example
"Symmetric"	left side is mirror image of right side	
"Positively skewed"	right tail is longer then the left	
"Negatively skewed"	left tail is longer than the right	
"Unimodal"	one highest point	
"Bimodal"	two high points	
"Normal"	unimodal, symmetric,	

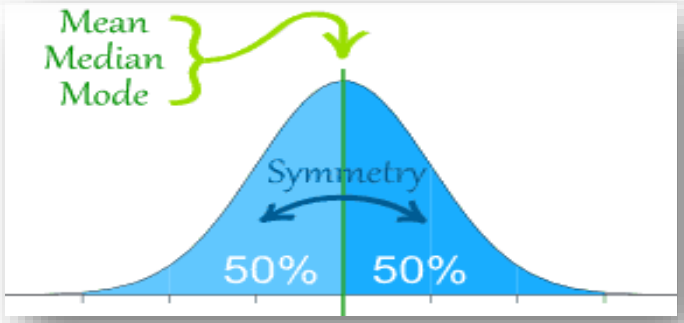
- Skewness
  - ✓ Positive skew
  - ✓ Negative skew
  - ✓ Normal distribution
  - ✓ **Quantile Definition of Skewness**

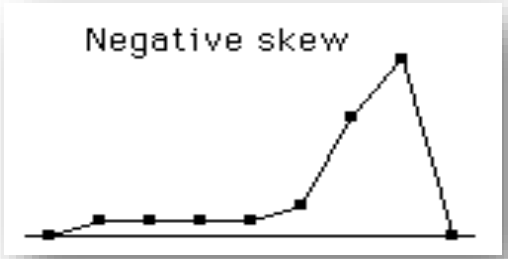
QUANTILE DEFINITION OF SKEWNESS

Type of Skewness	Quartile Relationship	Bowley's Skewness	Mean, Median, Mode
Symmetric (Normal)	Q3 – Q2 = Q2 – Q1	0	Mean = Median = Mode
Positive Skew	Q3 – Q2 > Q2 – Q1	> 0	Mean > Median > Mode Mean is greater than median
Negative Skew	Q3 – Q2 < Q2 – Q1	< 0	Mean < Median < Mode Mean is smaller than median

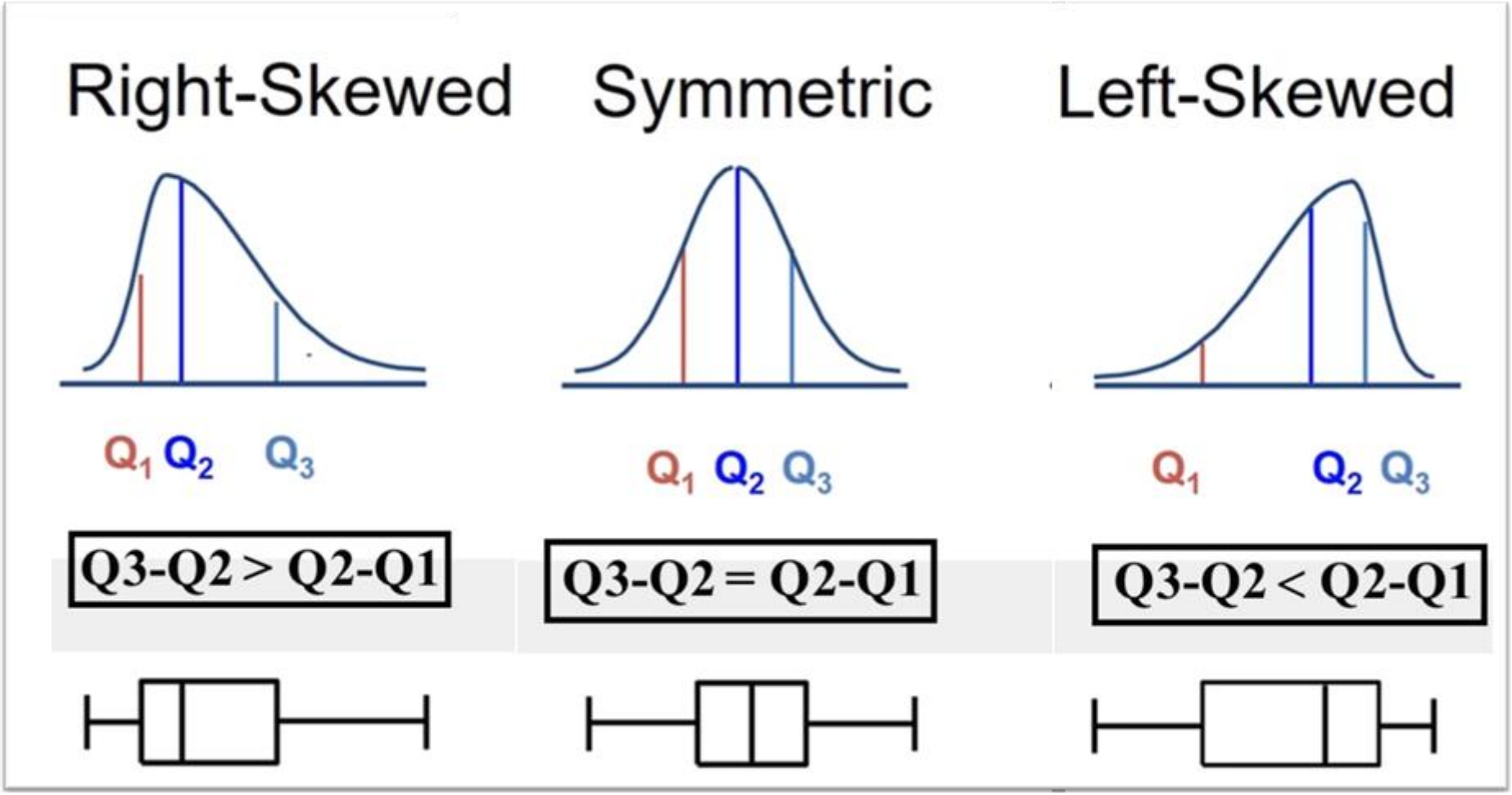
$$\text{Bowley's Skewness} = \frac{(Q_3 + Q_1 - 2Q_2)}{Q_3 - Q_1}$$







QUANTILE DEFINITION OF SKEWNESS



Right length (Q3-Q2) is larger than the left length (Q2-Q1)

Right length (Q3-Q2) is equal to the left length (Q2-Q1)

Right length (Q3-Q2) is smaller than the left length (Q2-Q1)

Find out whether the following data set is Zero Skewed, Positively Skewed, or Negatively Skewed using quartile values. Verify your answer by comparing the;

1. Quartile relationship
2. Mean, Median, and Mode
3. Bowley's Skewness Formula

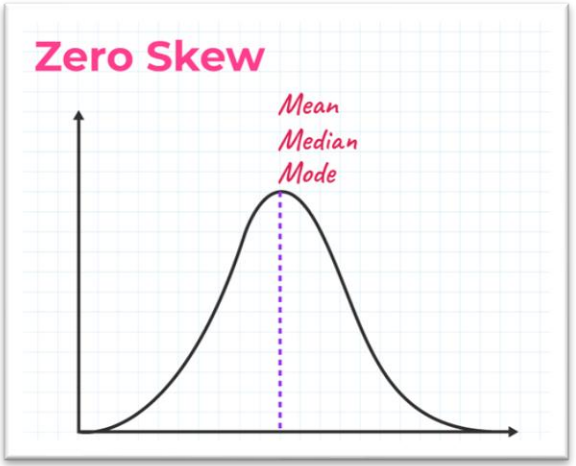
Data set 1	1, 3, 5, 5, 5, 7, 9
Data set 2	2, 7, 7, 8, 12, 14, 20
Data set 3	3, 6, 7, 9, 10, 10, 11

Remember this Rule تذكر هذه القاعدة

Type of Skewness	Quartile Relationship	Bowley's Skewness	Mean, Median, Mode
Symmetric (Normal)	Q3 – Q2 = Q2 – Q1	0	Mean = Median = Mode
Positive Skew	Q3 – Q2 > Q2 – Q1	> 0	Mean > Median > Mode Mean is greater than median
Negative Skew	Q3 – Q2 < Q2 – Q1	< 0	Mean < Median < Mode Mean is smaller than median

Data Set: 1 (1, 3, 5, 5, 5, 7, 9)

Q1= 4, Q2= 5, Q3= 6



Q3-Q2 = 6-5 = 1, Q2-Q1 = 5-4 = 1

Q3-Q2 = Q2-Q1
1 = 1 ⟶ ***Zero Skew***

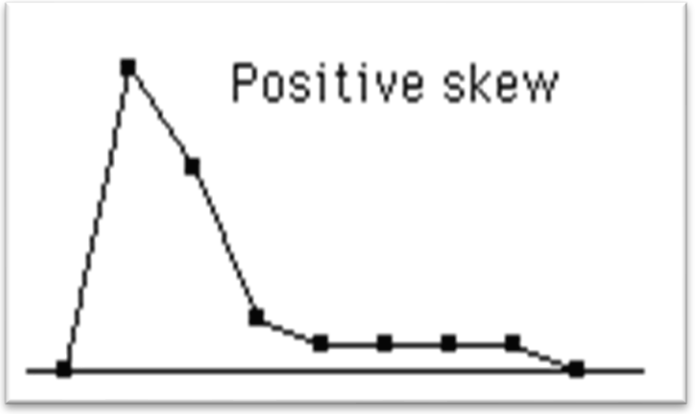
Mean = 5, Median = 5, Mode = 5

Mode = Median = Mean
5 = 5 = 5 ⟶ ***Zero Skew***

$$\text{Bowley's Skewness} = \frac{(Q_3 + Q_1 - 2Q_2)}{Q_3 - Q_1} = \frac{(6 + 4 - 2 \times 5)}{6 - 4} = \frac{(10 - 10)}{2} = \frac{0}{2} = 0$$

Zero Skew

Data Set: 2 2, 7, 7, 8, 12, 14, 20



Q1=7, Q2=8, Q3=14

Q3-Q2 = 14-8 = 6, Q2-Q1 = 8-7 = 1

Q3-Q2 > Q2-Q1 → Positive Skew

6 > 1

Mean = 10, Median = 8, Mode =7

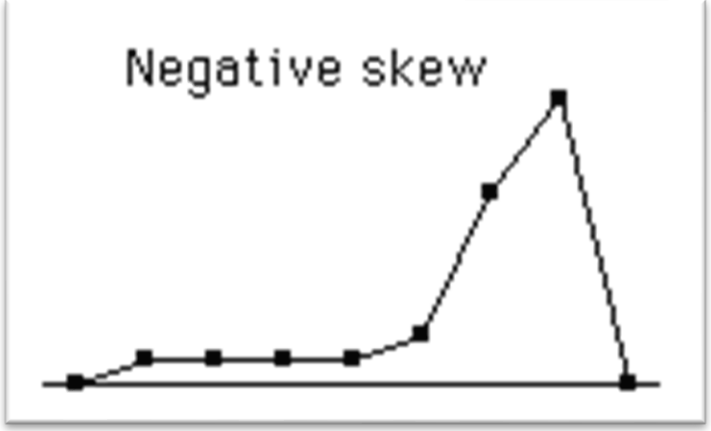
Mode < Median < Mean → Positive Skew

7 < 8 < 10

$$\text{Bowley's Skewness} = \frac{(Q_3 + Q_1 - 2Q_2)}{Q_3 - Q_1} = \frac{(14 + 7 - 2 \times 8)}{14 - 7} = \frac{(21 - 16)}{7} = \frac{5}{7} \approx 0.71$$

Positive Skew

Data set 3 (3, 6, 7, 9, 10, 10, 11)



Q1=6, Q2=9, Q3=10

Q3-Q2 = 10-9= 1, Q2-Q1 = 9-6 = 3

Q3-Q2 < Q2-Q1 → **Negative Skew**

1 < 3

Mean = 8, Median = 9, Mode = 10

Mode > Median > Mean → **Negative Skew**

10 > 9 > 8

$$\text{Bowley's Skewness} = \frac{(Q_3 + Q_1 - 2Q_2)}{Q_3 - Q_1} = \frac{(10 + 6 - 2 \times 9)}{10 - 6} = \frac{(16 - 18)}{4} = \frac{-2}{4} = -0.5$$

Negative Skew

THANK YOU